Merits of Geometric Mean:

- It is based on all the observations
- It is rigidly defined
- It is capable of further algebraic treatment
- It is less affected by the extreme values
- It is suitable for averaging ratios, percentages and rates.

Limitations of Geometric Mean:

- It is difficult to understand
- The geometric mean cannot be computed if any item in the series is negative or zero.
- The GM may not be the actual value of the series
- It brings out the property of the ratio of the change and not the absolute difference of change as the case in arithmetic mean.

5.3.3 Harmonic Mean (H.M.)

Harmonic Mean is defined as the reciprocal of the arithmetic mean of reciprocals of the observations.

(a) H.M. for Ungrouped data

Let $x_1, x_2, ..., x_n$ be the n observations then the harmonic mean is defined as

$$\text{H. M.} = \frac{n}{\sum_{i=l}^{n}\left(\frac{1}{x_i}\right)}$$

Example 5.11

A man travels from Jaipur to Agra by a car and takes 4 hours to cover the whole distance. In the first hour he travels at a speed of 50 km/hr, in the second hour his speed is 65 km/hr, in third hour his speed is 80 km/hr and in the fourth hour he travels at the speed of 55 km/hr. Find the average speed of the motorist.

Solution:

x	50	65	80	55	**Total**
$1/x$	0.0200	0.0154	0.0125	0.0182	**0.0661**

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